

The arch

An arch helps spread the weight of the part of a building that lies above it. The true arch was perfected by the Romans after 200 BCE, and allowed larger and lighter buildings, while using less stone or brick.

Corbel arch

The first arches, such as this Gate of the Lions at Mycenae, Greece (c 1250 BCE), were corbelled. This means they were built with layers of stone, each jutting out further until they met at the top.

The design did not spread weight evenly and corbel arches needed lintels (horizontal blocks for support) below, or reinforcement at the sides.



Triumphal arch

Having mastered the true arch, the Romans built longer bridges, created aqueducts to carry water, and raised domed buildings by using an extended arch as a roof. They put up triumphal arches to celebrate the victories of their emperors. The Arch of Titus (c 82 CE) in Rome is one of the most splendid.

Concrete

The Romans discovered concrete around 200 BCE when they found that adding lime to pozzolana, a type of sand found near Rome, made it harden quickly. Buildings made with concrete needed less stone, which was expensive. Roman architects used the new material for constructing enormous buildings such as the Colosseum (72–80 CE) and the Pantheon (118–125 CE).

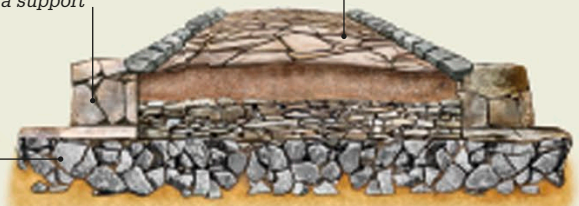
Colosseum, Rome



Kerbstones at the side of the ditch give extra support

Finer sand and concrete form the top layer

Layer of larger stones and rubble fills the ditch



Cross-section of a Roman road

Road construction

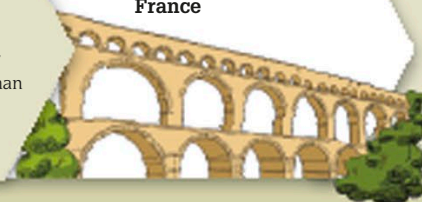
The Romans were excellent engineers, and built a large network of high-quality roads to link towns within their empire. To make a road, a ditch was dug and filled with layers of rubble, then smaller stones, and finally fine sand and concrete on top. The most important roads were then surfaced with cobbles.

The 80 concrete arches on each story strengthened the building and allowed crowds of spectators to enter easily.

c 60 CE

The Pont du Gard is one of the greatest Roman aqueducts, built to carry water into the Roman town of Nemausus (modern Nîmes). More than 900 ft (275 m) long, it originally had 60 concrete arches on three levels.

Pont du Gard, France



c 126 CE

The Pantheon was built in Rome by Emperor Hadrian. Its enormous dome, 141 ft (43 m) high, is still the largest unsupported concrete dome in the world.

683 CE

The Temple of the Inscriptions was completed at Palenque, Mexico. A monument to its Mayan ruler K'inich Janaab' Pakal, it is the largest pyramid structure in Central America.